

Activity-2

1. Differentiate between Statement, PreparedStatement, and CallableStatement. Compare them with respect to syntax, performance, security, and practical use cases.
2. What is connection pooling? Why is it required in enterprise database applications? Explain its working principle and advantages over normal JDBC connections.
3. A web application connects to the database thousands of times per minute. Explain why DriverManager-based connections may become inefficient and how DataSource with connection pooling solves the problem.
4. Explain with examples how SQL query results are mapped into Java objects. Discuss the role of getters like getInt(), getString(), getDate(), and the importance of correct type mapping.
5. Compare java.sql API and javax.sql API in detail. Highlight their major features, differences, and practical usage in enterprise applications.
6. Evaluate the role of javax.sql in modern enterprise database programming. Why is it considered an improvement over basic JDBC usage?